Chapter Three
The Topography of Ethiopia and the Horn

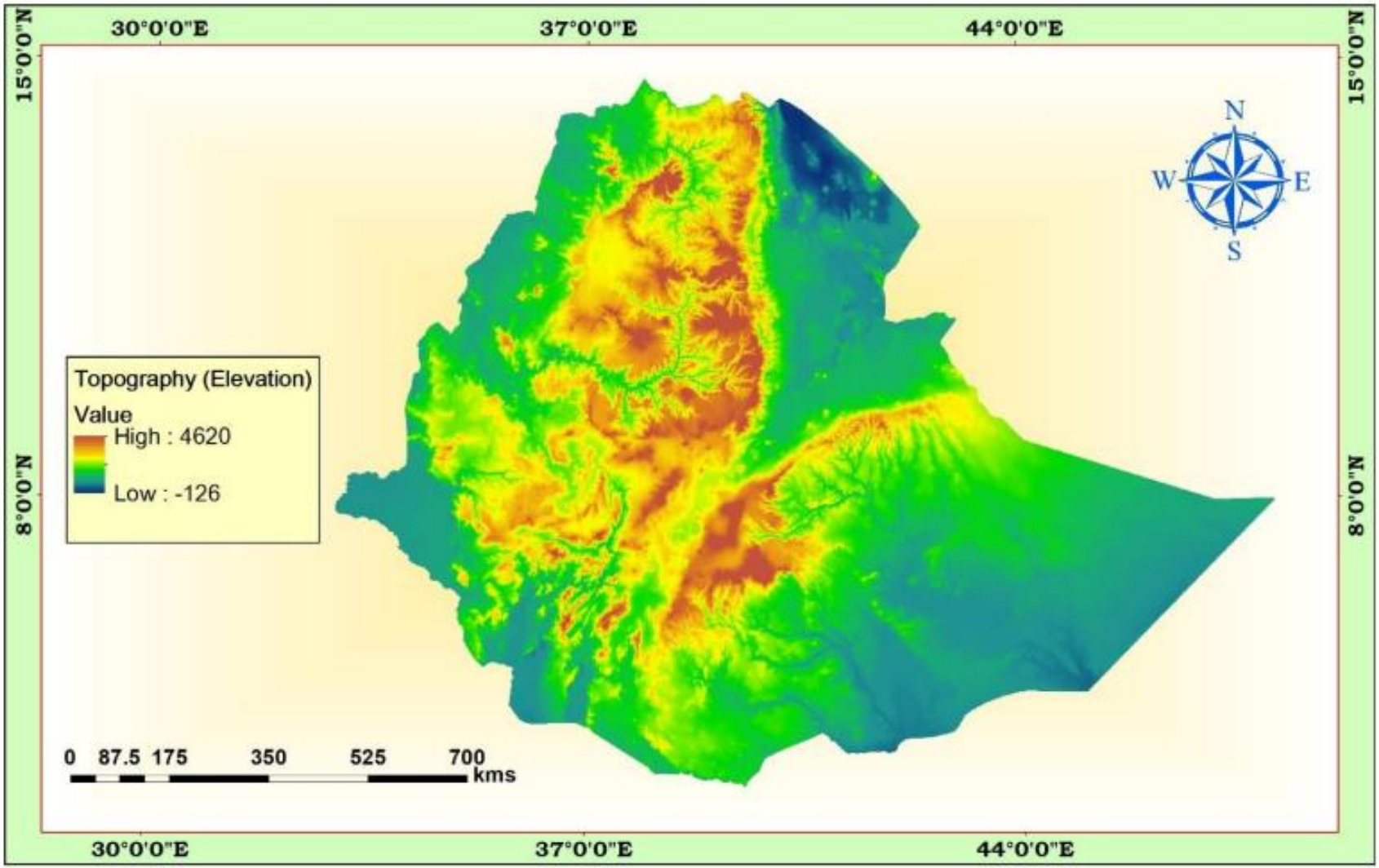


Figure 3.1: Ethiopian Topography

Source: CSA shapefile, 2012

Objectives

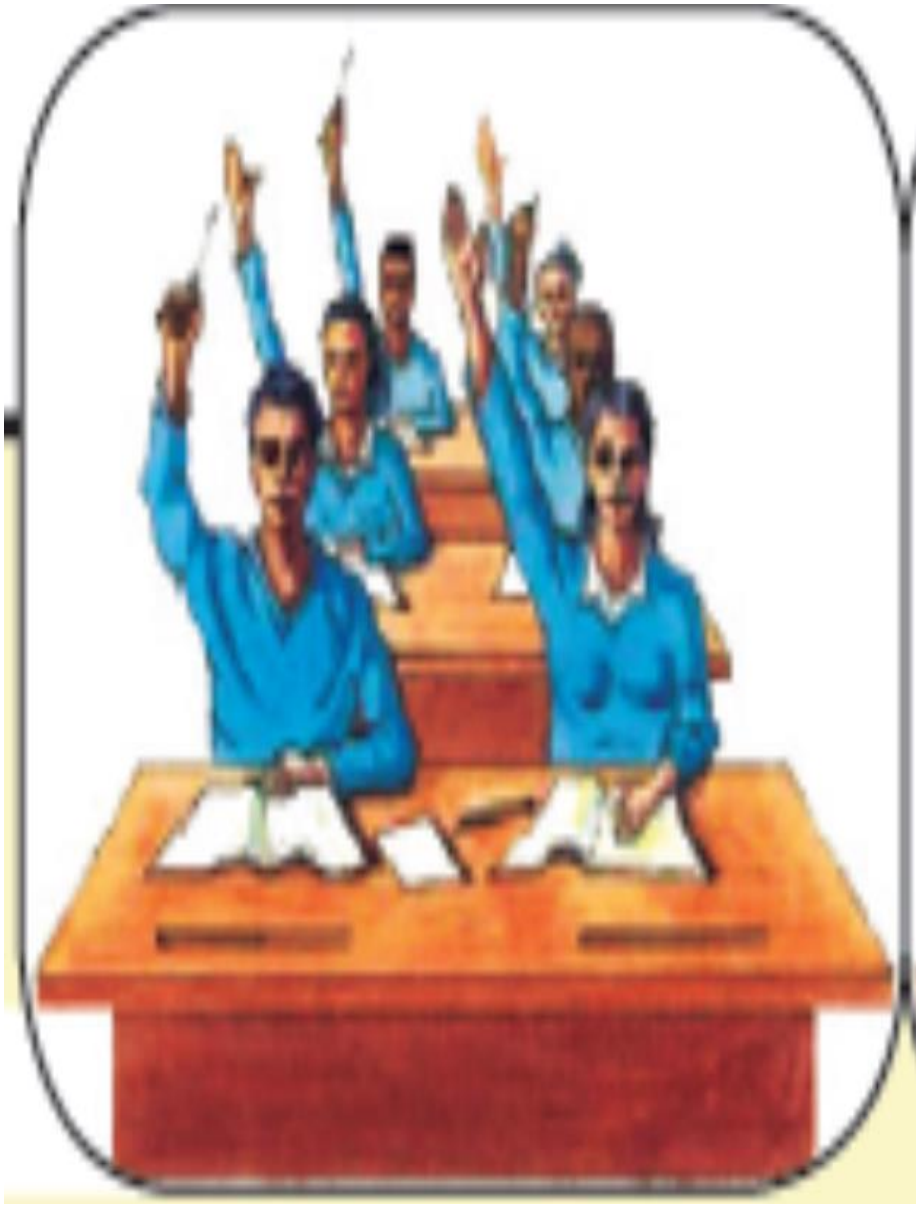
At the end of this chapter, you should be able to:

- Describe the topography of Ethiopia and the Horn.
- Identify the physiographic divisions of Ethiopia.
- Explain the physiographic characteristics of the Rift Valley.
- Explain the impacts of relief on biophysical and socioeconomic conditions

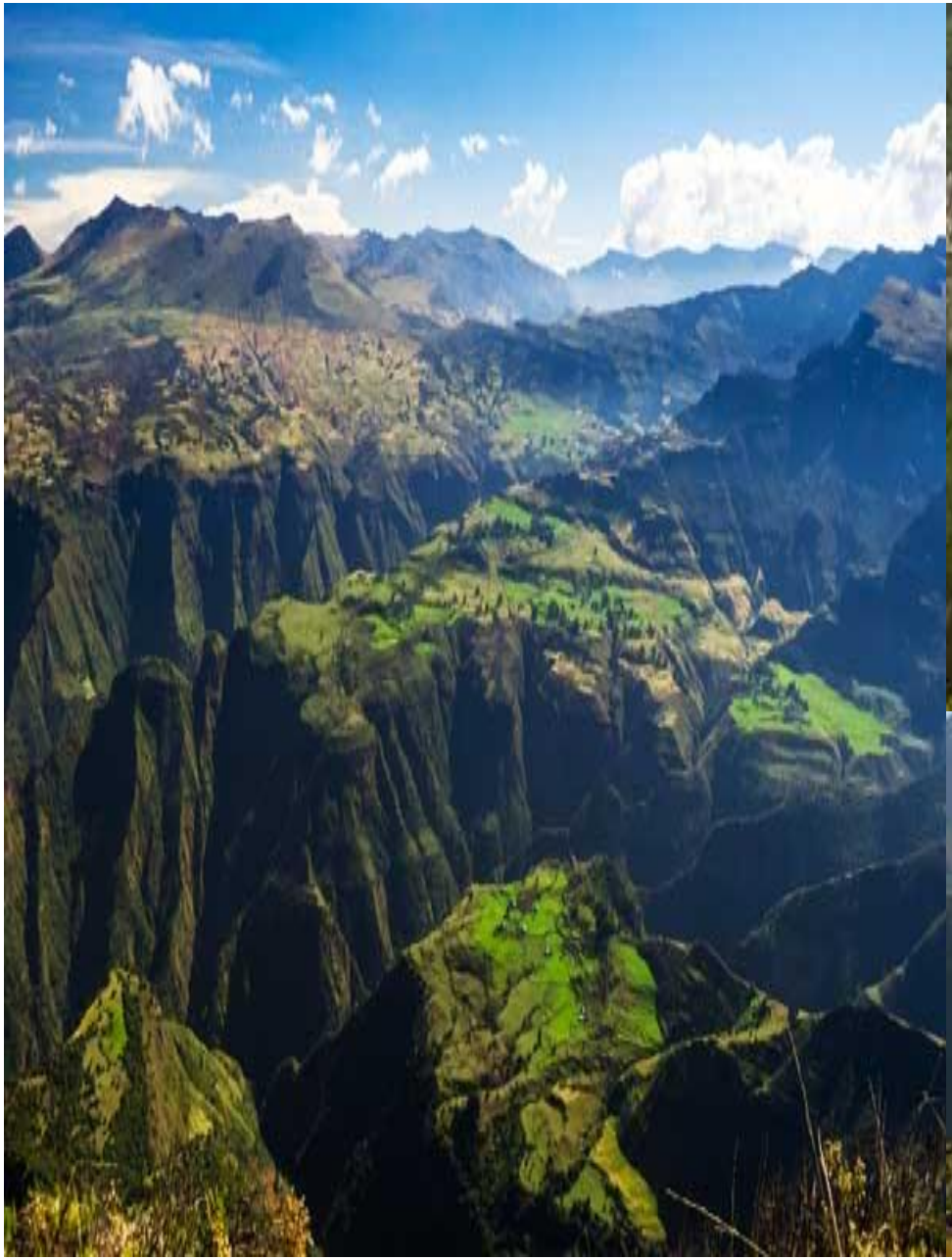
TOPOGRAPHY OF ETHIOPIA AND THE HORN…

Brainstorming questions

- ❑ What is topography?
- ❑ What is the
  - Environmental
  - Socio-economic,
  - Political implications of Ethiopian topography?
- ➢ What expression of '***Ethiopia is a roof of East Africa***' **implies?**



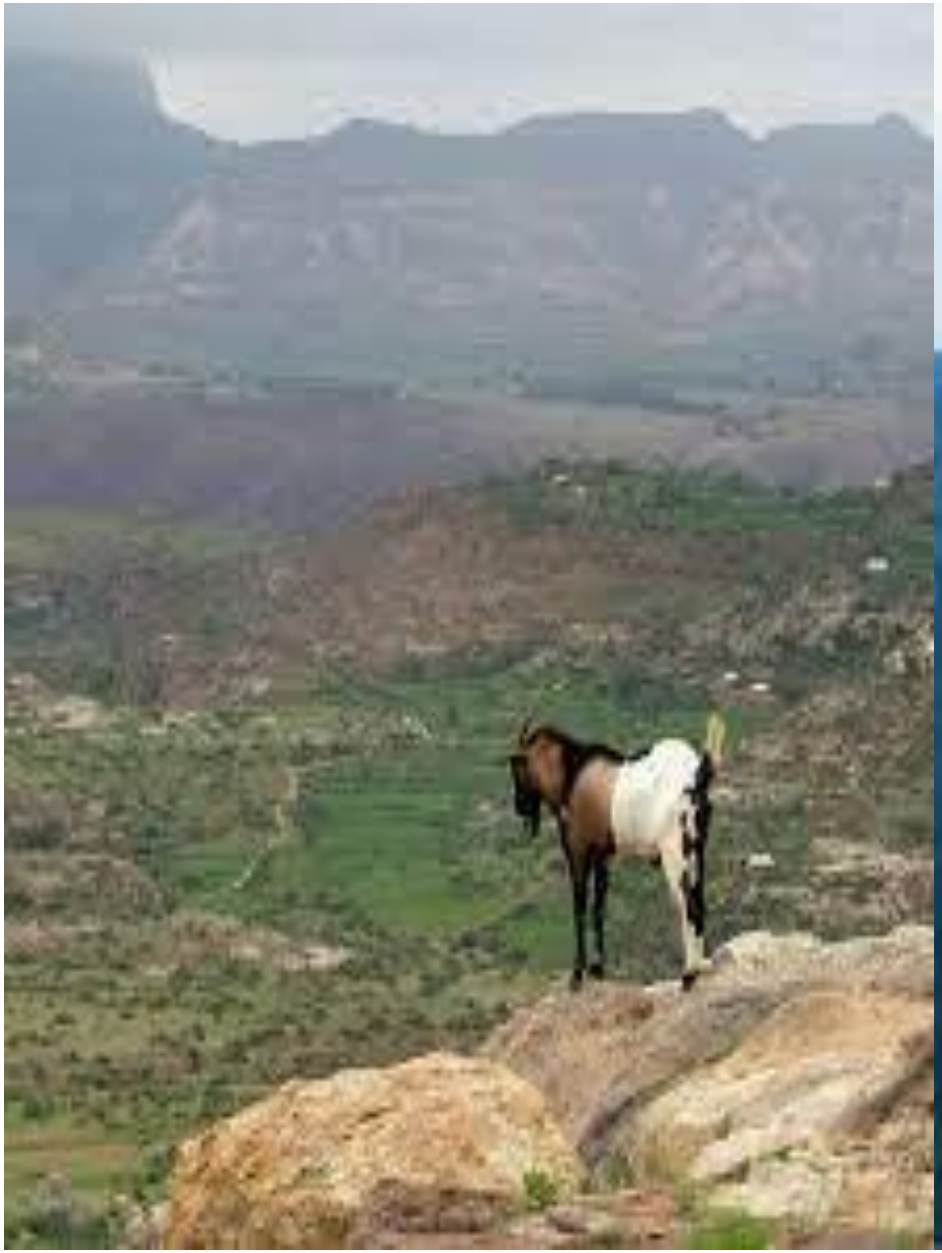
Peculiar example







Peculiar example…





2.1. General Characteristics of the Ethiopian Physiography







- The term **physiography** is also called **relief** or **landform** or **terrain**.
- It is **a natural** or **artificial feature** of the solid surface of the Earth.
- Both by **African** & **world standards** Ethiopia is a country characterized by **topographic diversity of high** & **rugged mountains**, **flat topped plateaus**, **rolling plains**, & **deep gorges** as well as river valleys.

General Characteristics of Ethiopian Physiography

- Ethiopian landform is characterized by great diversity:
  - Plateaus,
  - High and rugged mountains,



- Deep river gorges and……
- Vast plains.



Con't……





- Altitude ranges from

125 mbsl(Kobar Sink) to

4,620 masl (Mt. Ras Dashen).

- Roof of East Africa
- Using the 1000 masl - highlands & lowlands

Cont'd…

- Diversity in topography brought differences in natural features;
  - soil,
  - climate,
  - vegetation and
  - wild life &
  - socio-cultural and economic phenomena.
- 1000 m contour is the demarcation line b/n **highlands & lowlands.**

Con't……

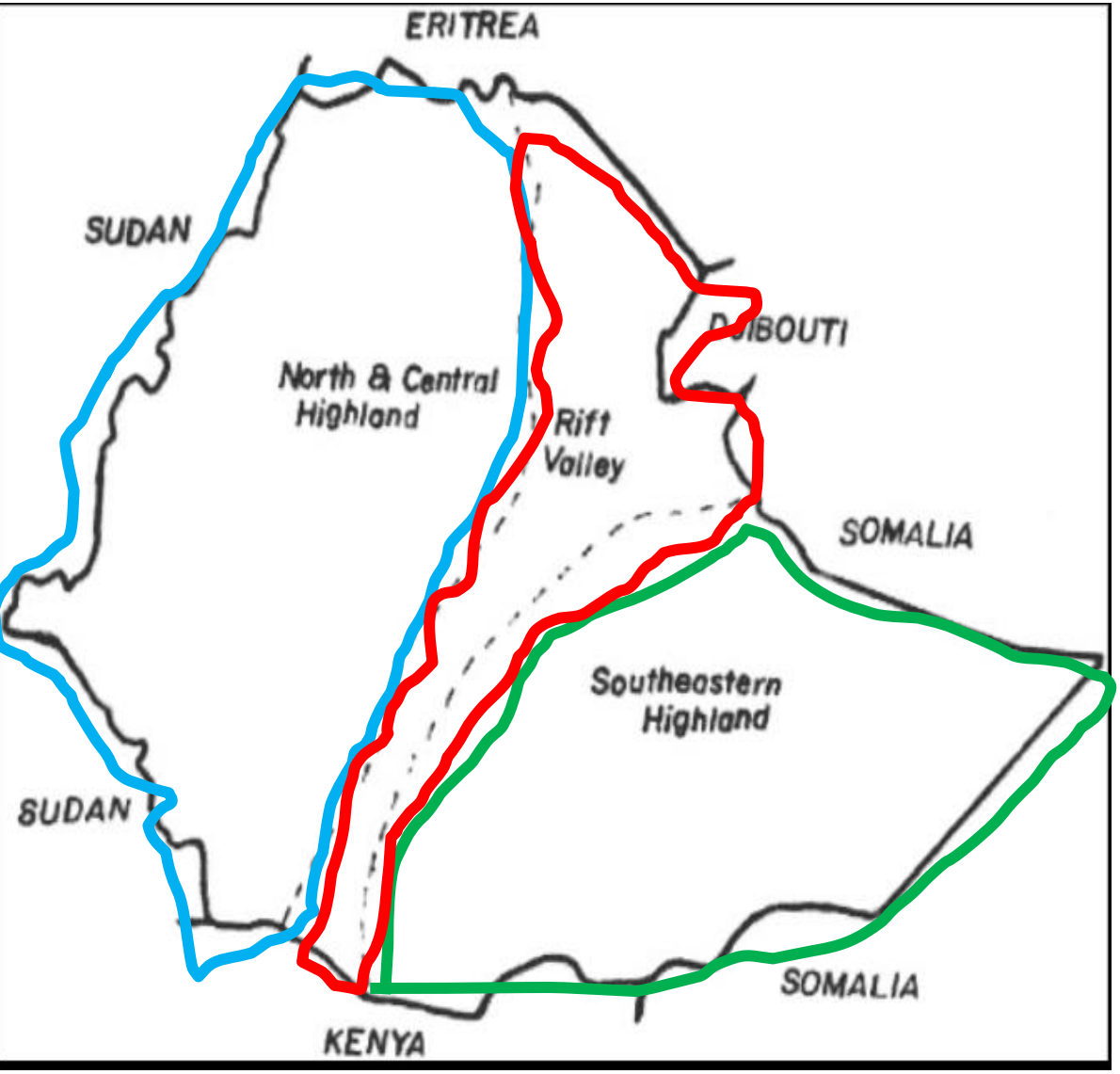
The Highland Areas of Ethiopia

- Areas having greater than **1000 masl**
- 56% of Ethiopia's total area
- Characteristics of Ethiopian highlands:
  - ✓ Moderate and high amount of rainfall (>600 mm per year).
  - ✓ Lower mean annual temperature (<20^0C).
  - ✓ The climate is favourable for biotic life.
  - ✓ Rain-fed agriculture is possible.
  - ✓ Free from tropical diseases.
  - ✓ Attractive for human habitation and densely settled.

The Lowland Areas of Ethiopia

- Low relief areas with less than 1000 meters altitude.
- Covers 44% of the country's area
- Ethiopian lowlands are characterized by:
  - ✓ Fewer amounts of rainfall and higher temperature.
  - ✓ High prevalence of tropical diseases.
  - ✓ Lower population densities.
  - ✓ Pastoral and semi- Pastoral economic life.
  - ✓ Vast plain lands favourable for irrigation agriculture along the lower river basins.

3.2. The Physiographic Divisions of Ethiopia



The formation of rift valley

1. The Western highlands and lowlands
2. The South-eastern (Eastern) highlands and lowlands
3. The Rift Valley

- Each division has its own peculiar features.

3.2.1. The Western Highlands and Lowlands

- Include all areas west of the Rift Valley.
- It extends from **north to south** encompassing nearly the **whole western half** of Ethiopia.
- the **biggest region** covering **44% of the area** of the country.



This region is further subdivided into;

1. **Western Highlands** (76.3%) and
2. **Western Lowlands** (23.7%).

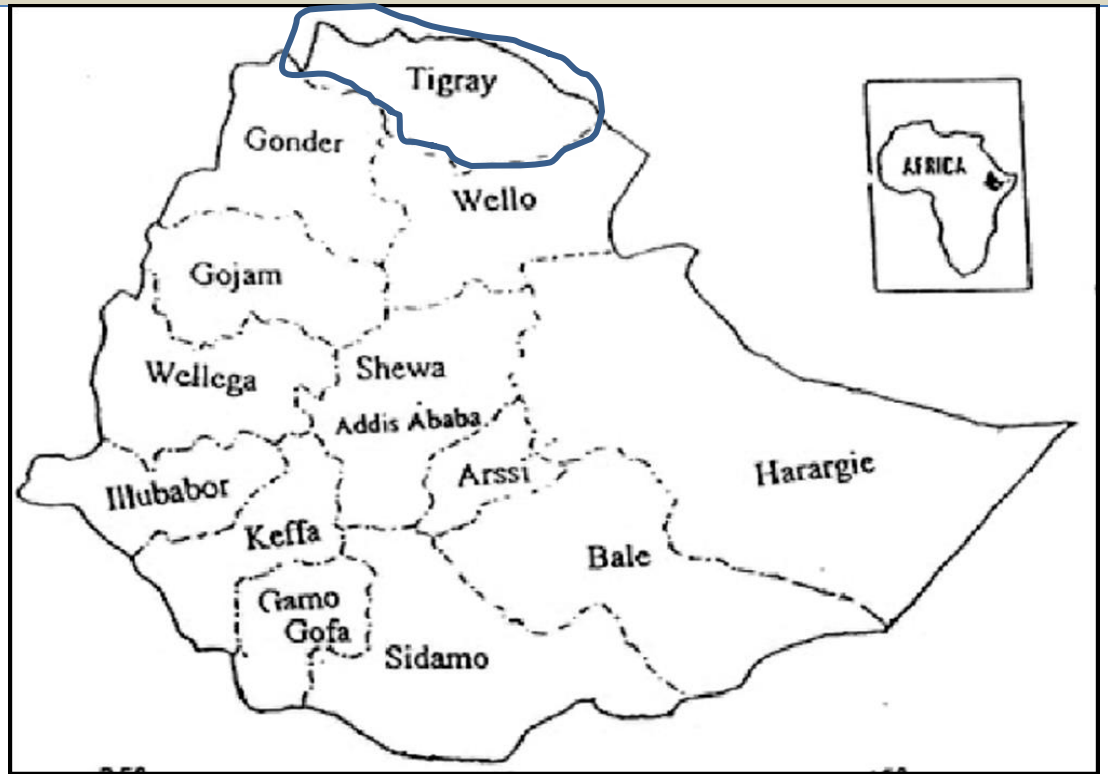
- Both regions has **four groups** of highlands and **four groups** of lowlands

1. The Western Highlands

- Tigrean plateaus
- North Central Massifs
- Shoan plateaus
- South western highlands

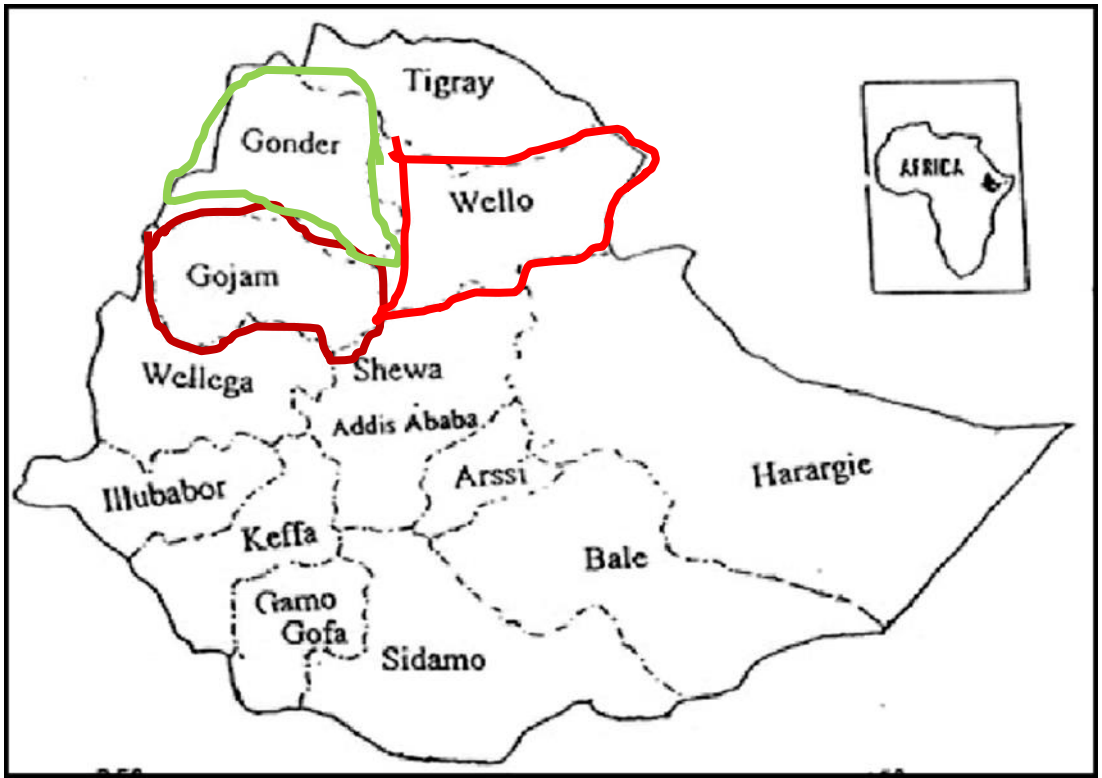
Tigrean plateaus

- Is the most northerly & smallest plateau region in Ethiopia.
- Is drained & dissected by Tekkezze & Mereb rivers.
- Mt. Ambalaghe (3291 m) & Mt. Asimba (3248 m) are the highest peaks.



North - Central Massifs

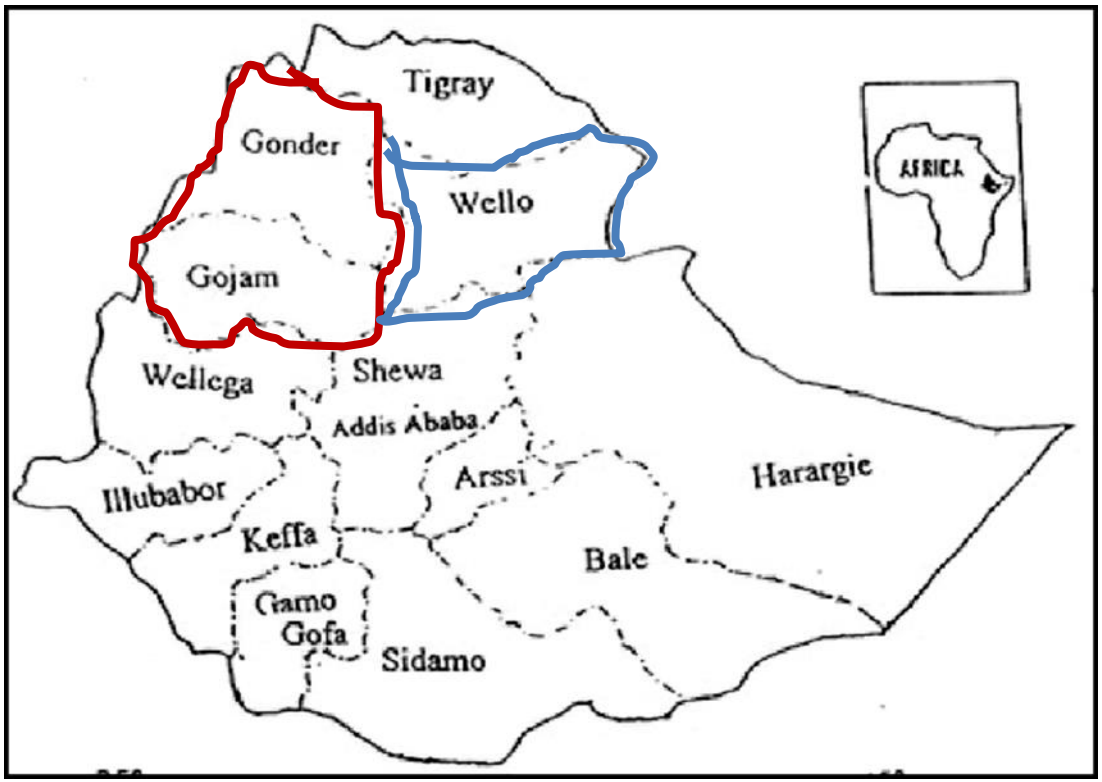
- Located to the south of the Tigrean plateaus, it is found between the Tekkezze & Abay (Blue Nile) rivers.
- It is the biggest by area among the divisions of Western highlands
- Contains 18 of the 26 highest mts. in Ethiopia, of which Ras Dashen (4620 m) is the highest of all.
- comprises the geographical territories of Gondar, Wollo & Gojjam.



- Divided by the gorges Abay & Tekkezze massively, it has two broad sections; namely

1. **Western Group** consisting Gojjam & Gondar
2. **Eastern Group** holding Wollo

❑ Yeju-wadla Delanta ridge is the only higher ground connecting these two regions.



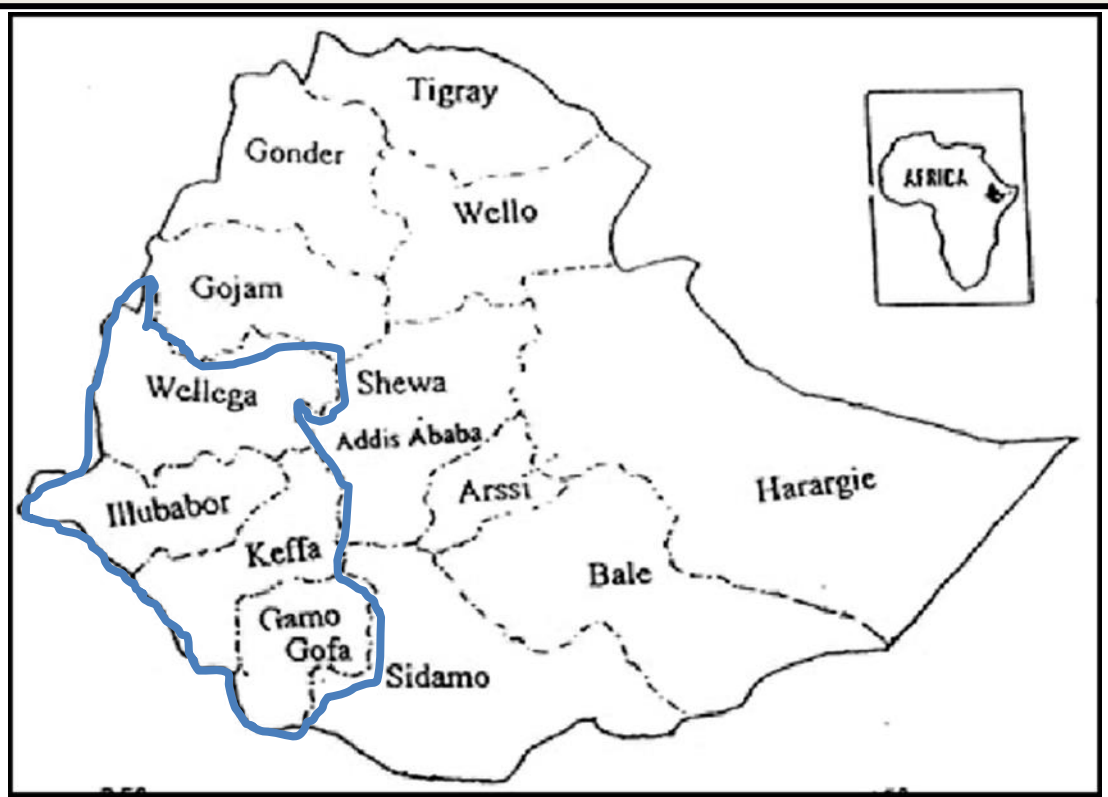
Shoan plateaus

- Also known as Central highlands, it has a central position in the country.
- It has the **most elevated landmass** in Ethiopia, because about 75% of its total area is located at more than 2000 meters a. s. l.
- It creates a **watershed** among Awash, Abay & Ghibe rivers.
- Mt. Abuye Meda (4000 meters) is the highest summit in the region.



Southwestern highlands

- Represents the **higher parts** of
  - ✓ Wellega,
  - ✓ Kaffa,
  - ✓ Illubabor & Gamo Gofa.
- Drained by several streams which carry large amount of water.
- Thus, it is the **most physically dissected** plateau region in Ethiopia.
- Is the **most rain fed region** in Ethiopia.



- Is the **most vegetated region**, partly because of abundant water supply.
- Is the **major coffee producing region** in Ethiopia
- Guge Mountain (Arba Minch/ Gamo highlands), (4200 m) in the is the highest summit in the region



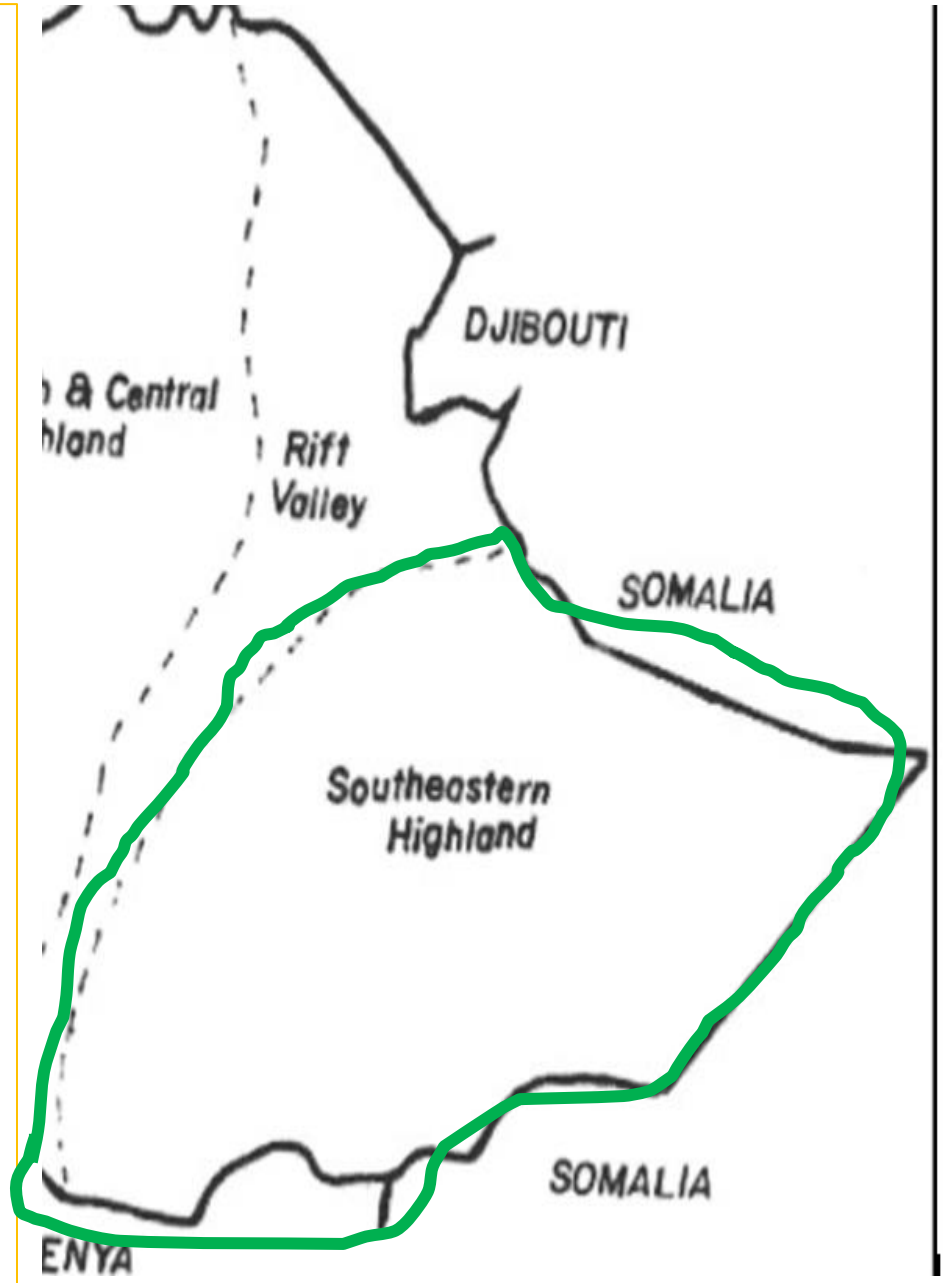
2. The Western Lowlands

1. Tekkezze-Anghereb lowlands
2. Abay-Dider lowlands
3. Baro (Gambella) lowlands
4. Ghibe lowlands

- Baro lowlands are **Ethiopia's wettest lowlands** largely because of their favorable location in terms of the Southwest Guinea Monsoon winds which bring big volume of rain water during the **Kiremt (Summer) season**.
- Thus, as we go from these lowlands within the region, rainfall decreases, while temperature increases.
- As a result in most parts of the region **pastoral economic activities** become common practice.

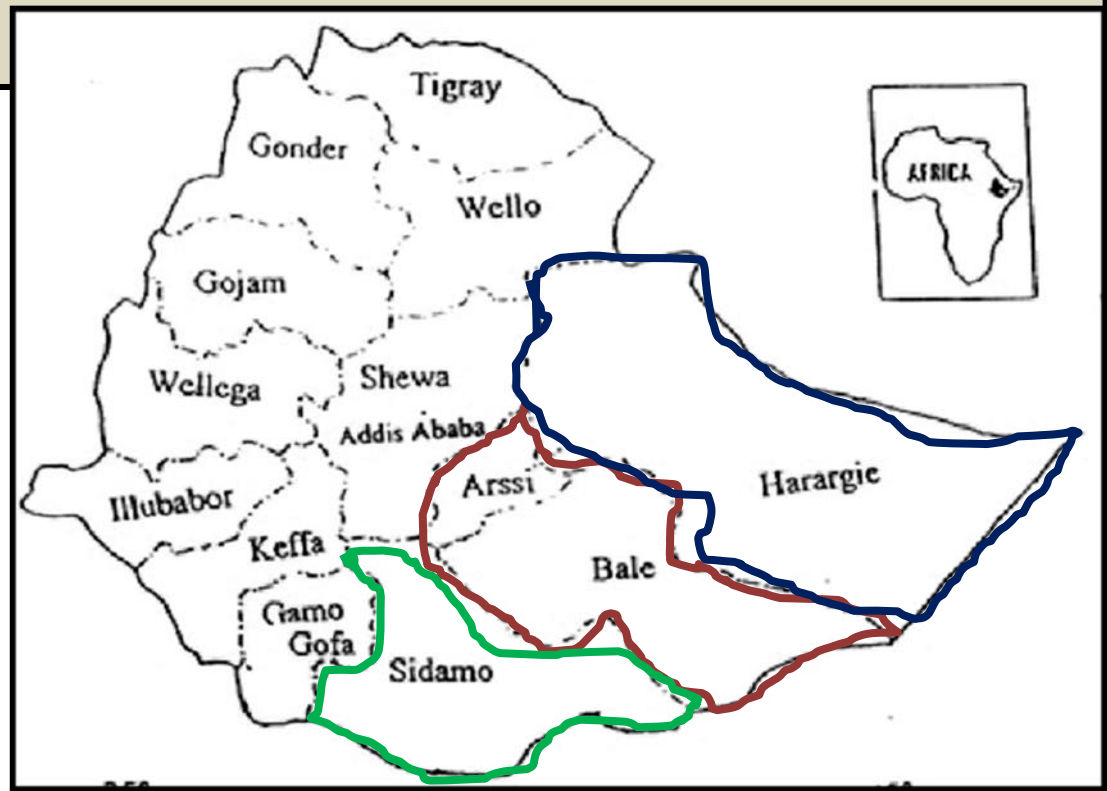
3.2.2. The Southeastern Highlands and Lowlands

- Located to the east of the Rift valley
- is the 2nd largest covering 37% of the country's total area.
- Covers the geographical territories of the Harari & Somali regions fully & that of Oromia & SNNPR regions partially
- Is classified into: Southeastern highlands & Southeastern lowlands



1. The South-eastern Highlands

- Makes up 46% of the physiographic division
- It has three divisions
    1. The Arsi-Bale Highlands
    2. Sidamo Plateaus
    3. The Hararghe Plateau



The Arsi-Bale Highlands

- The **biggest by area**, **highest by altitude** & have **a central location** in the region.
- Combine higher areas of Arssie & Bale.

- Highlands in Bale & Arssie are **separated** by the headstream of Wabishebele river, Wabe which is one of its tributaries.
- Arssie-Bale highlands are known for **cool weather crops** production such as wheat & barley.
- Major peaks in Arssie include;

i. Mt Chilalo (4036 m) ii). Mt. Bada (4139 m)

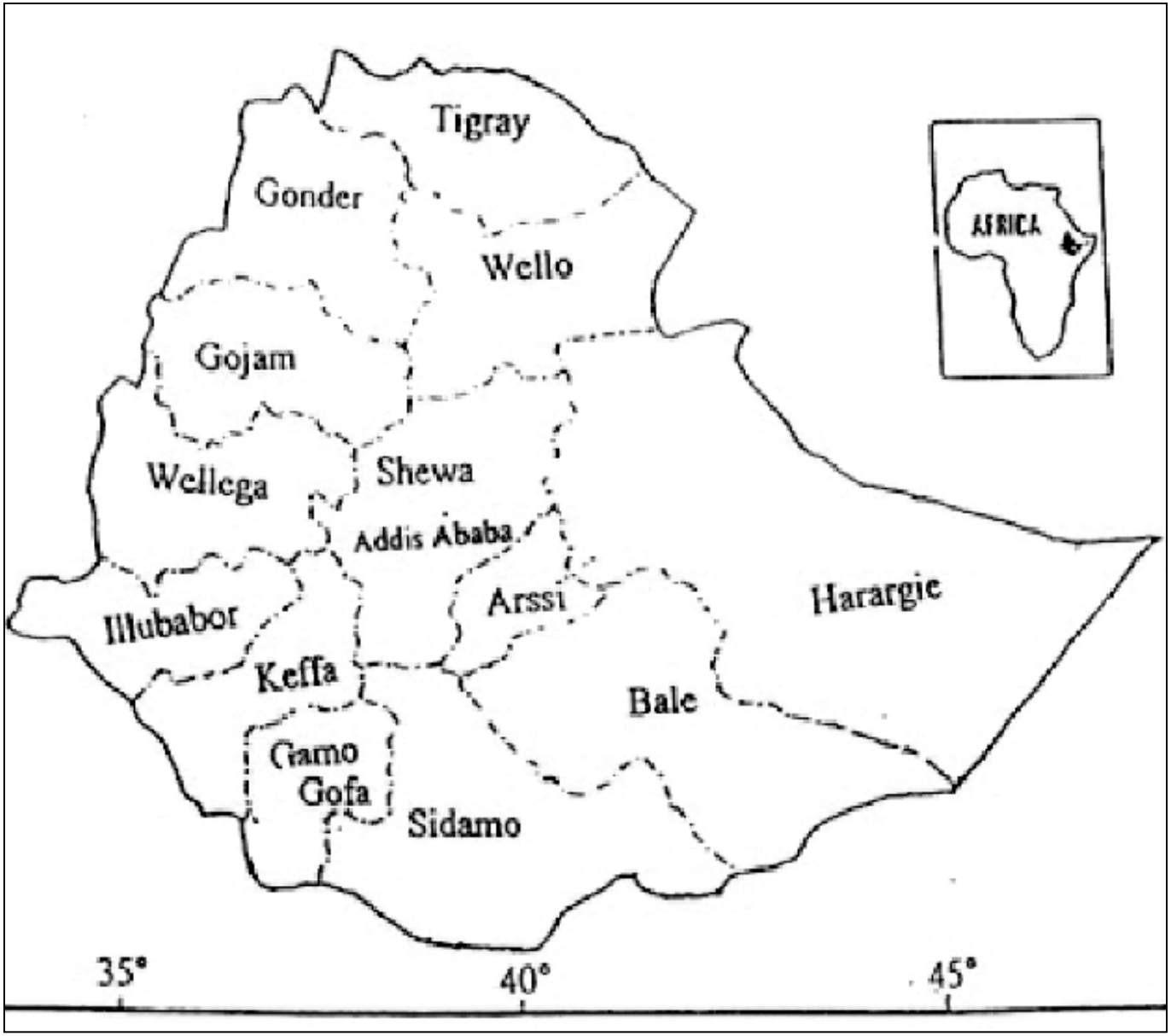
iii). Mt. Kaka (4180 m) iv). Mt. Kara (4340 m)

- Major peaks in Bale include;

i. Mt. Batu (4307 m)

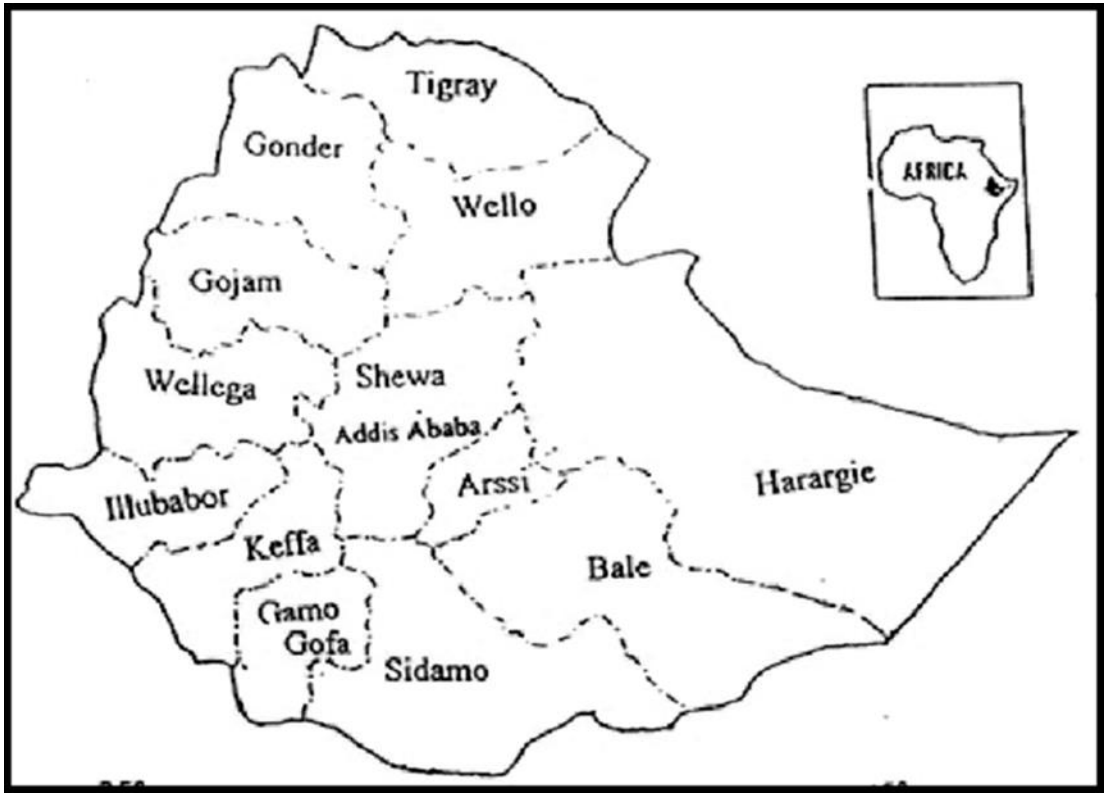
ii. ii). Mt. Tulu Deemtu (4377 m).

Con't……..


Tigray
Gonder
Wello
Gojam
Wellega
Shewa
Addis Ababa
Illubabor
Arssi
Harargie
Keffa
Bale
Gamo
Gofa
Sidamo
35°
40°
45°
AFRICA

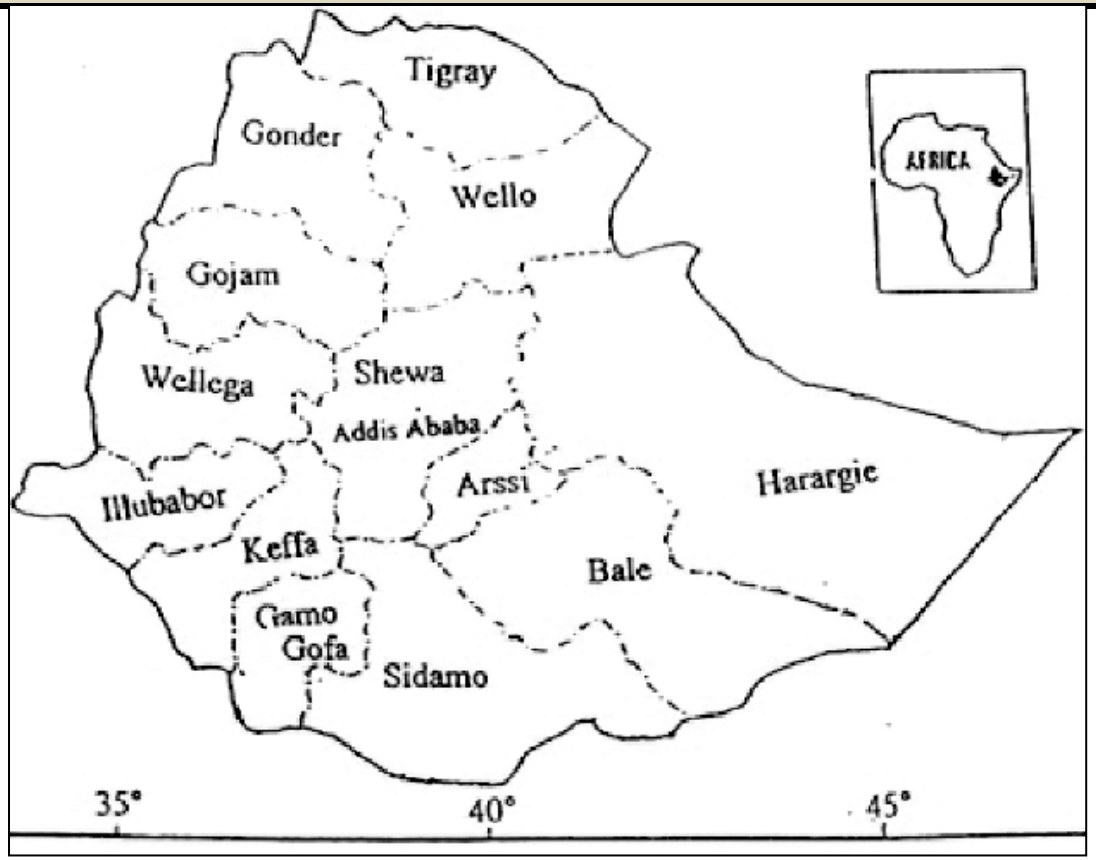
Sidamo Plateaus

- These are located to the **southwest** of the Arssi-Balle massifs drained by river Dawa & other tributaries of Genale river.
- It is well known **for coffee production** & **gold mines**
- Mt. **Dello** (3600 m) is the highest summit here.



The Hararghe Plateau

- It is the **northeast extension** of the Arssi-Bale massifs which is popular for its best quality coffee type produced in Ethiopia, **coffee Arabica**.
- Mt. Garamuleta (3381 m) is the highest peak.



2. The South-eastern Lowlands

- Makes up 54% of the physiographic region
- Located along the Ethio-Somalian & Ethio-Kenyan borders, are the **most extensive lowlands**

❑ Cover more than a quarter of Ethiopia's total area .

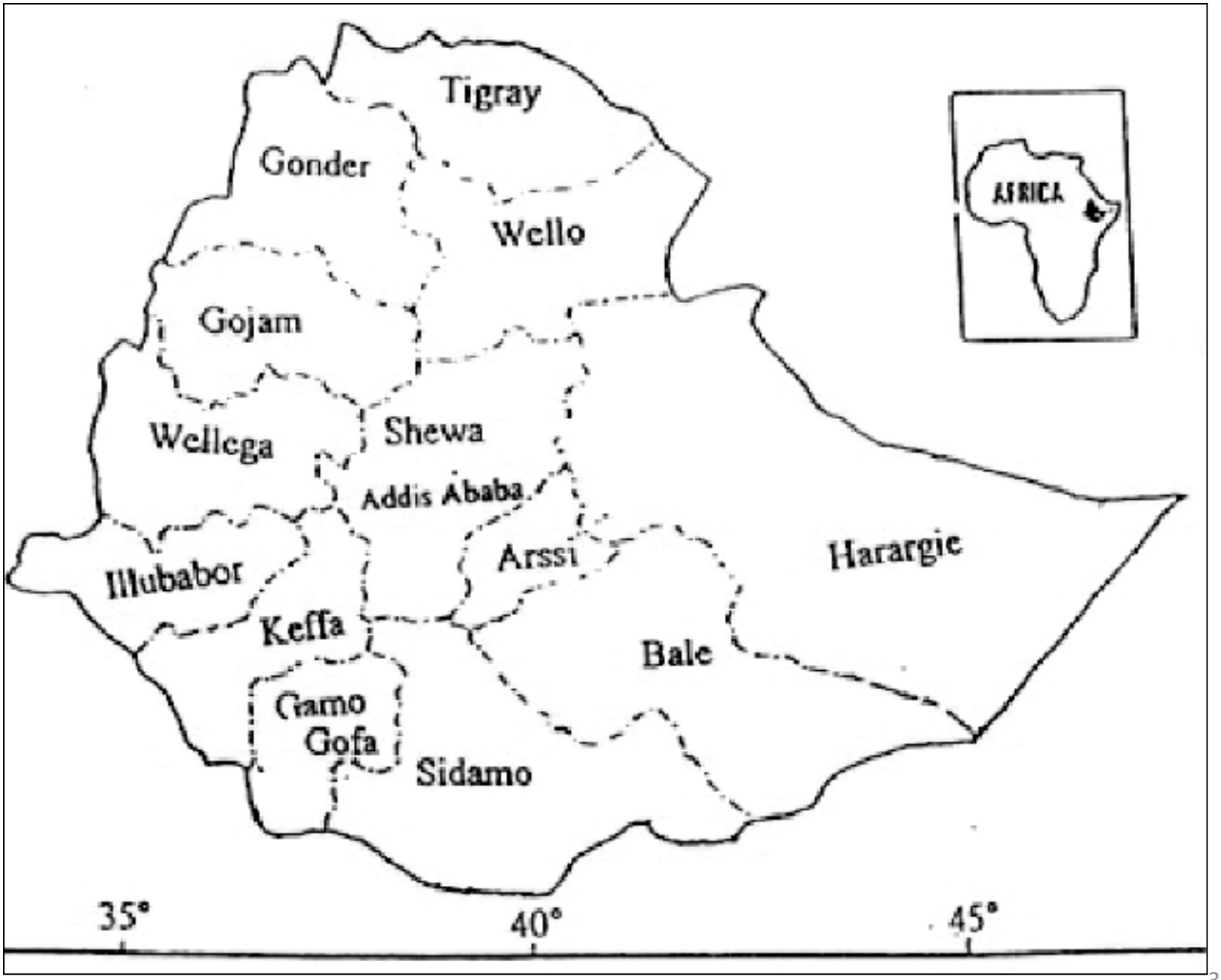
❑ They have two main divisions named after the two principal rivers draining the region.

A. The **Shebelle lowlands** which encompass the lowlands of Ogaden & Elkere

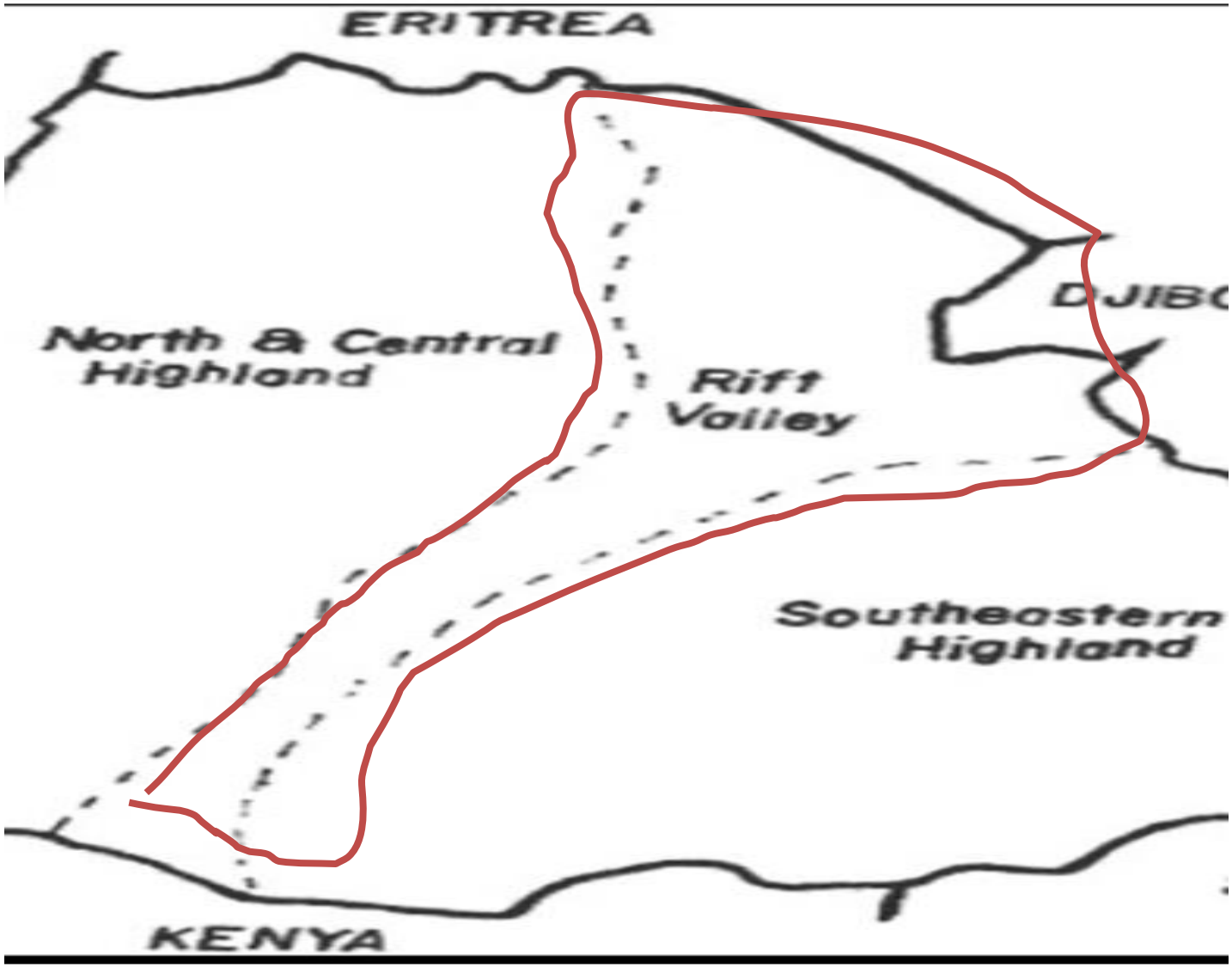
B. The **Genale lowlands** made up of lowlands of Borena.

❑ Due to scarcity of rainfall, **Pastoral economic** culture is common in most parts of the region.

❑ The region is known for its big potentials of **petroleum, natural gas** & **large cattle population**.

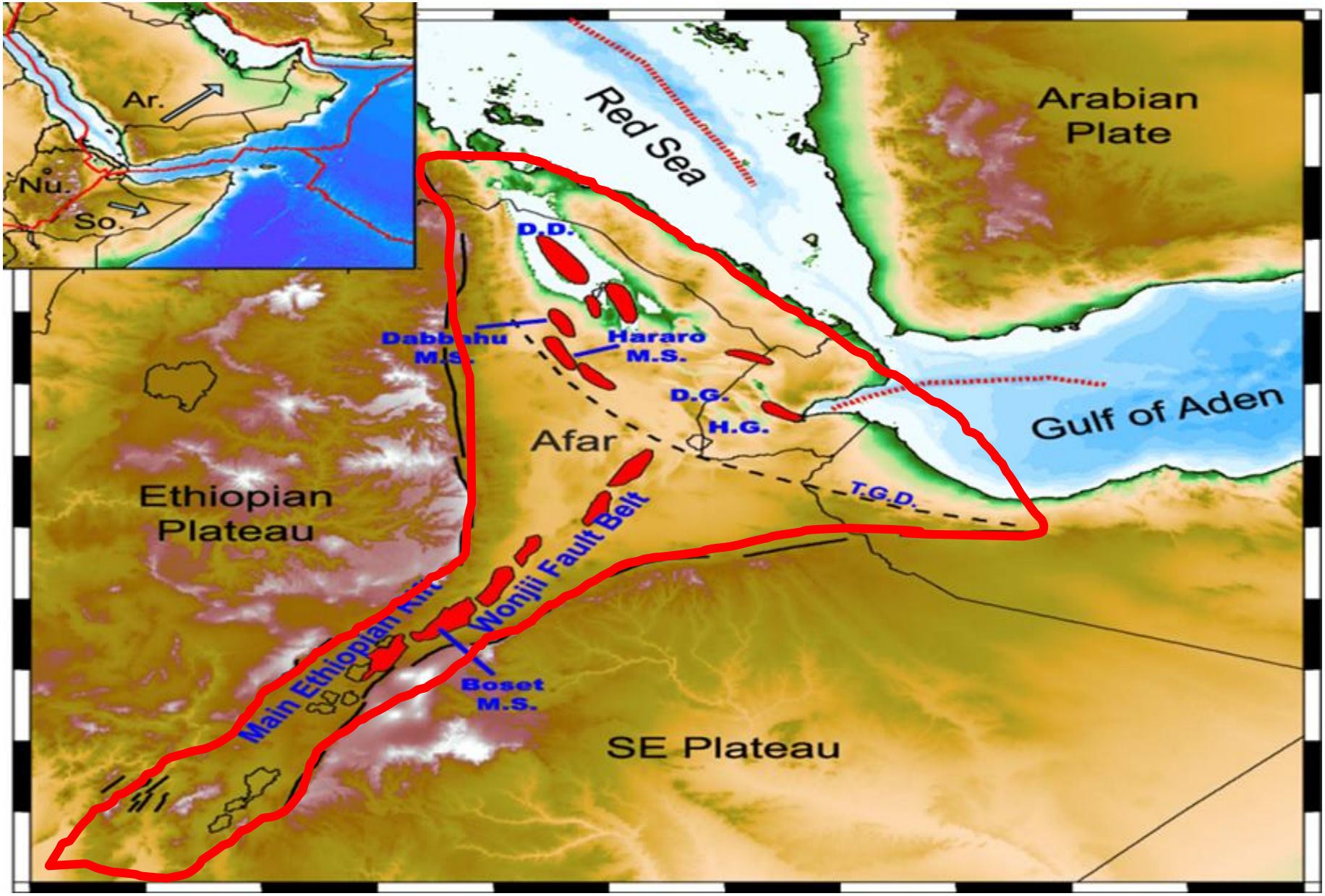

Tigray
Gonder
Wello
Gojam
Wellega
Shewa
Addis Ababa
Arssi
Harargie
Illubabor
Keffa
Bale
Gamo
Gofa
Sidamo
AFRICA
35°
40°
45°

RIFT VALLEY


ERITREA
North & Central Highland
Rift Valley
Southeastern Highland
KENYA

3.The Rift Valley

- It is a structural depression formed by the tectonic force
- The formation of the Rift Valley has separated the Ethiopian Highlands and Lowlands in to two
- It extends from the Afar triangle in the north to Chew Bahir for about 1,700 km
- It covers 18% of the area of Ethiopia
- It is elongated with a NE-SW orientation


Ar.
Nu.
So.
Red Sea
Arabian Plate
D.D.
Dabbahu M.S.
Hararo M.S.
D.G.
H.G.
Afar
Gulf of Aden
T.G.D.
Ethiopian Plateau
Wonjii Fault Belt
Main Ethiopian Rift
Boset M.S.
SE Plateau

Rift Valley is further subdivided into **three physiographic** sub-regions

1. The Afar Triangle: characteristics

- is the largest and widest part of the Rift Valley
- 54% of the Rift Valley area
- Morphologically, it has triangular-shape lowland (1,000 m to 125 mbsl
- One of the most hostile (**Very unfavorable to life**) environments on Earth (maximum temperatures can exceed 50°C during the summer wet season.

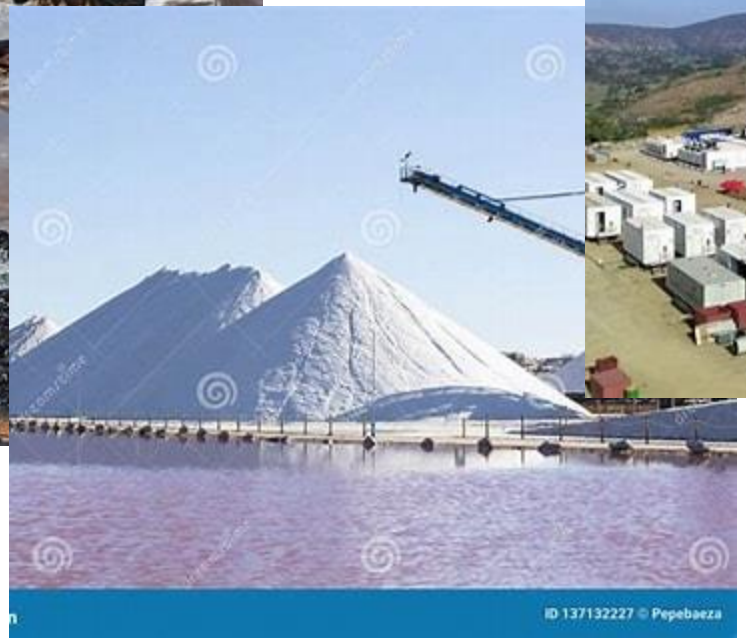
- Located at low altitude b/n (300-700m) & characterized by:
  - ☞ faulted depressions (grabens), volcanic hills, active volcanoes, volcanic ridges, lava fields and low lava platforms.
  - ☞ Have Lakes like Abe, Asale, and Afrera and Denakil Depression (Kobar Sink).
  - ☞ Generally hot and dry



The economic importance of the region:

- ❖Salt extraction,
- ❖irrigation along the Awash River and electric potential from geothermal energy







2. The Main Ethiopian Rift/ Central Rift

- It refers to the **narrow belt** of the Rift Valley that extends from **Awash River** in the north to **Lake Chamo** in the south.
- It has an average width of 50-80 km and elevation of 1,000-2,000 masl

The presence of cinder cones and volcanic mountains like:

- Mount Fentale,
- Boseti-guda (near Adama),
- Aletu (north of Lake Ziway) and Chebi (north of Lake Hawasa).

Other resource base of the region:

- recreational value of the lakes,
- the agriculture and the geothermal energy potential

- The lakes of the region is generally milder and watery.

The presence of cinder cones and volcanic mountains like:

- Mount Fentale,
- Boseti-guda (near Adama),
- Aletu (north of Lake Ziway) and Chebi (north of Lake Hawasa).

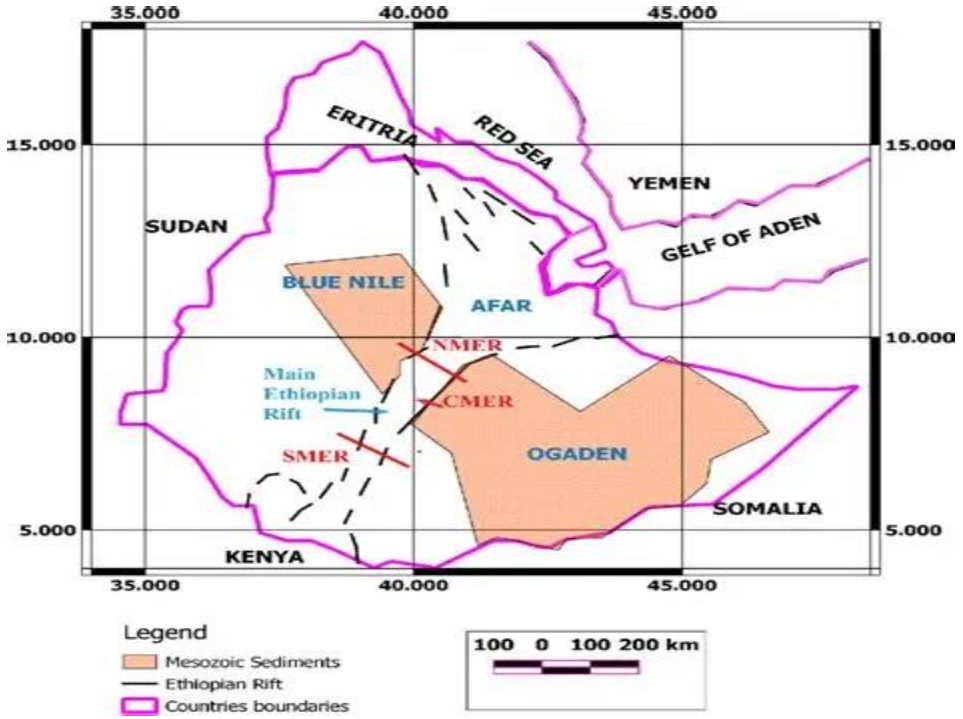




3. **The Chew Bahir Rift**

- Is the smallest part of the Rift Valley that includes the Gneissic highlands of Konso
- Is the **broad** and **shallow** depression.
- It includes the **streams** of **Segen** and **Woito.**








gettyimages
Credit: REDA&CO

3.3. The Impacts of Relief on Biophysical and Socioeconomic Conditions





Impacts of Relief on Biophysical and Socioeconomic Conditions

- Relief influence the various socioeconomic aspects of Ethiopia

- ***Relief influences on Agricultural practices***:
  - Choice of farming techniques (e.g. in rugged terrain mechanized farming techniques are difficult to practice).
- ***Relief influences crop production*; e.g**
- Corps-barley, wheat are well adapted to higher altitudes whereas
- Crops -sorghum, maize are well adapted to *lower altitude*.

- ***Relief influences the practices of animal husbandry*** :
  - equines and sheep are reared in the **higher altitudes** and
  - camels and goat are well adapted to **lower altitudes**
- ***Influences on Settlement pattern****:*
  - *Highlands* of Ethiopia that experience a temperate type of climatic condition that are mainly free from most of the tropical diseases are densely settled.
- The highlands are characterized by **sedentary life** and permanent settlements whereas,
- lowlands are inhabited by **pastoralists** with temporary settlements.

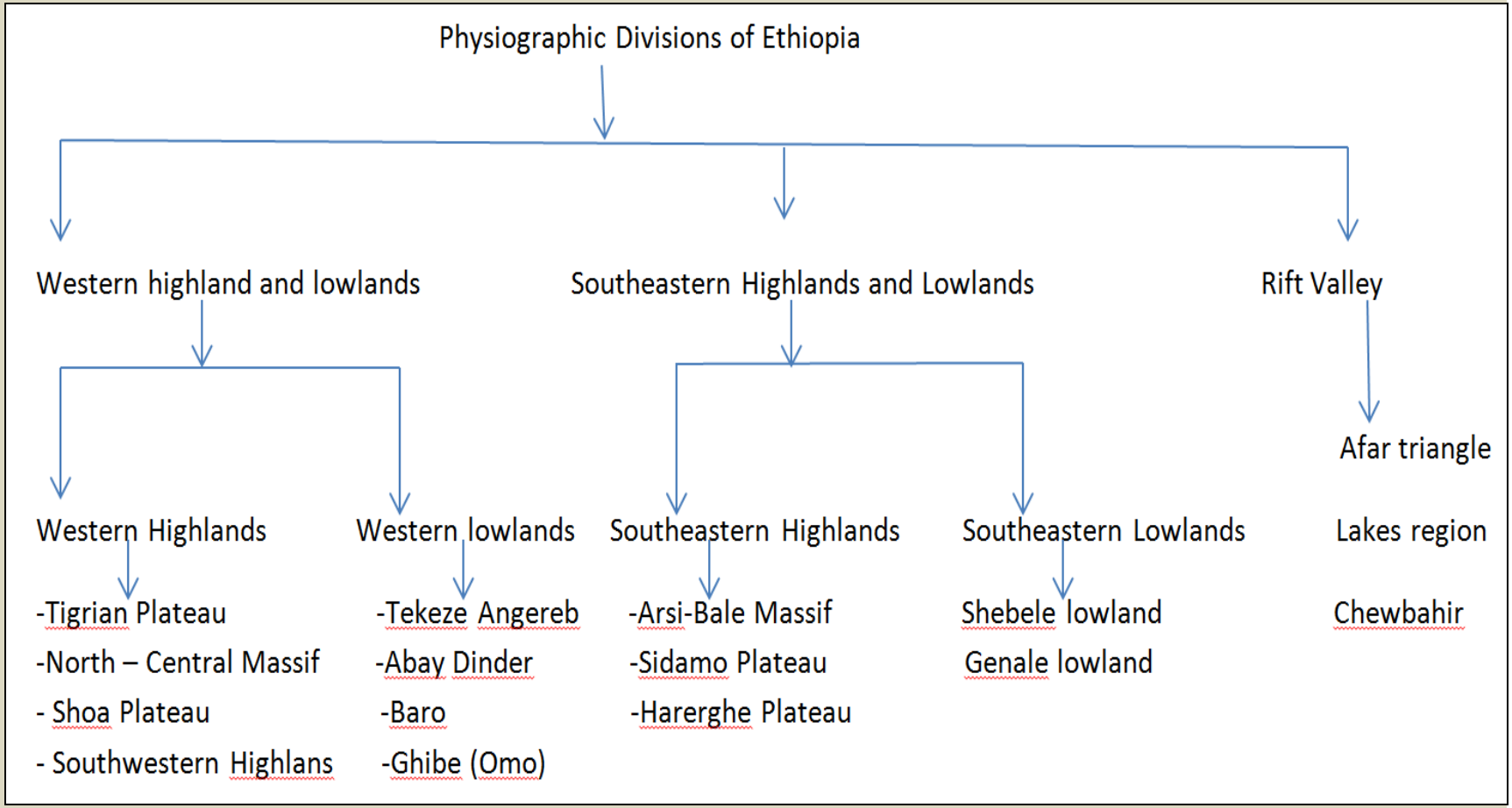
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- Influences on Transportation and communication
- Hydroelectric power potential
- Socio-cultural feeling
- Impacts on climate
- Impacts on soil
- Impacts on natural vegetation

Summary of the Lesson

Main points of the lesson

- General Characteristics of Ethiopian topography



- Impacts of Relief